



# Paula Sánchez Checa

PhD student

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## Links

🆔 ORCID

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## About

Paula Sánchez Checa is a second-year PhD student in the Computer Science and Engineering Department at Universidad Carlos III de Madrid. Her research combines high-performance computing and epidemiological modelling, and she is currently working on epidemiological modelling using artificial intelligence techniques. In the field of

computing, her research focuses on system modelling using AI-based tools and mathematical models, with the aim of combining the dynamic reconfiguration of resources of malleable applications and ad-hoc file systems in HPC environments. In the field of epidemiology, she has experience working with the EpiGraph epidemiological simulator and has collaborated with the European Centre for Disease Prevention and Control (ECDC) by participating in the ECDC Respiratory Diseases Scenario Modelling Hub (RespiCompass), providing predictions of epidemiological and vaccination scenarios for RSV. In addition, she collaborates with Centro Nacional de Epidemiología (Spanish National Centre for Epidemiology - CNE) modelling adverse weather events and infectious diseases in Spain.

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## Publications

*No recent journal publications found.*

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## Projects

**Modelado de eventos meteorológicos adversos y enfermedades transmisibles en España (MET).** (Modelling of adverse weather events and transmissible diseases in Spain (MET)).

**Nuevas técnicas escalables de E/S para cargas de trabajo híbridas de HPC e intensivas en datos. - SCIOT.** (New scalable I/O techniques for hybrid HPC and data-intensive workloads. - SCIOT). [SCIOT web page](#) 🔗

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## Teaching

### Graduate teaching assistant

#### Operating Systems

Spring Term 2026. Bachelor's Degree in Computer Science and Engineering. Second year subject. 6 ECTS. Delivered the course entirely in English.

### Computer Architecture

Autumn Term 2025. Bachelor's Degree in Mathematics and Computing. Third year subject. 6 ECTS. Delivered the course entirely in English.

Autumn Term 2025. Bachelor's Degree in Computer Science and Engineering. Third year subject. 6 ECTS. Delivered the course entirely in Spanish

### Distributed Systems

Spring Term 2025. Bachelor's Degree in Computer Science and Engineering. Third year subject. 6 ECTS. Delivered the course entirely in Spanish.

Spring Term 2025. Double Degree in Computer Science and Engineering and Business Administration and Management. Fifth year subject. 6 ECTS. Delivered the course entirely in Spanish.

## Collaboration and Community Activities

Member of the Dynamic Resources for HPC Consortium (DynResHPC). [DynResHPC web page](#) 

Participation in the RespiCompass (ECDC Respiratory Diseases Scenario Modelling Hub) 2025 RSV Scenario Round. [RespiCompass web page](#) 

Student volunteer & registration assistant in the 30th International European Conference on Parallel and Distributed Computing (Euro-Par 2024).

## Talks and Presentations

**Comparative Analysis of Algorithms for Malleability Decision-Making in Applications and File Systems** (2025) 4th EuroHPC Workshop on Dynamic Resources in HPC at the 31st International European Conference on Parallel and Distributed Computing (Euro-Par 2025).

**Explorando la maleabilidad combinada de aplicaciones y sistemas de ficheros** (2025) XXXV Jornadas de Paralelismo (JP2025) at Jornadas SARTECO 2025.

**A Novel Prediction for Combined Computational and Storage Malleability** (2024) 3rd International EUGAIN Summer Training School: Gender & Sustainability.

## Posters

**Managing Malleability in Applications and File Systems** (2025) 3rd CINI HPC Summer School.

**A Novel Prediction Model for Combined Computational and Storage Malleability** (2024) 11th ACM Celebration of Women in Computing (womENCourage 2024).

## Student Projects

Roldán Matea, Manuel (2025/2026) **Desarrollo de algoritmos basados en inteligencia artificial para la reconfiguración dinámica de aplicaciones maleables.** (*Development of artificial intelligence-based algorithms for the dynamic reconfiguration of malleable applications.*) [Bachelor's Thesis]

Stefan Giurca, Adrián (2025/2026) **Detección de anomalías en tiempo de ejecución en entornos distribuidos.** (*Detection of runtime anomalies in distributed environments.*) [Bachelor's Thesis]

Navarro Hurtado, Pablo (2025/2026) **Gestión dinámica de recursos de cómputo y E/S: caso de estudio con EpiGraph y Expand Ad-Hoc.** (*Dynamic management of computing and I/O resources: a case study featuring EpiGraph and Expand Ad-Hoc.*) [Bachelor's Thesis]

## Conferences

**XXXVI Jornadas de Paralelismo (JP26)** - Madrid, June 2026. [JP26 web page](#) 